Analysis of Capital Asset Pricing Model (CAPM)

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Abstract

This study examines the CAPM using S&P 500 stock data from 2000–2026 by estimating beta coefficients and testing their ability to explain expected returns. While betas are statistically significant and vary across assets, the relationship between beta and returns is weak and sometimes inconsistent. Portfolios are constructed using equal weighting, which may introduce bias and affect the results. Overall, the findings provide only limited support for CAPM, suggesting that beta alone does not fully explain cross-sectional differences in returns.

1 Data Collection and Estimation of Beta Coefficients

1.1 Data Loading and Preparation

We use a dataset containing historical daily stock price data for 472 current S&P 500 companies, covering the period from January 2000 to February 2026. This dataset is one of the most comprehensive freely available S&P 500 datasets on Kaggle, consisting of approximately 2.7 million observations over 26 years of U.S. equity market history. The dataset is available at: https://www.kaggle.com/datasets/jacksaleeby/s-and-p500-historical-data. To ensure sufficient time series length for reliable beta estimation, we retain only stocks with at least 60 monthly observations. After applying this filter, the dataset is reduced from 472 stocks to 461 stocks.

Table 1: Summary statistics of monthly stock returns

Mean	SD	Min	Q1	Median	Q3	Max
0.0093	0.0983	-1.8546	-0.0368	0.0131	0.0598	1.524

Monthly returns are computed from end-of-month prices as logarithic returns:

$$r_{j,t} = \ln\left(\frac{P_{j,t}}{P_{j,t-1}}\right) \tag{1}$$

As the risk-free rate, we use the 3-Month Treasury Bill Secondary Market Rate (Discount Basis), which is a commonly used proxy for the risk-free rate in econometric studies. The data are obtained from the Federal Reserve Economic Data (FRED) database: https://fred.stlouisfed.org/series/DTB3.

The series is available at a daily frequency and in annualized percentage terms. It is converted to monthly frequency by taking the last observation of each month and transforming it into a monthly rate:

$$r_{f,t} = \frac{r_{f,t}^{(annual)}}{100 \cdot 12}. \tag{2}$$

This transformation provides an approximation of the effective monthly risk-free rate.

1.2 Visualization of Time Series

The time series in Figure 1 shows the evolution of the market return and the risk-free rate over time. The risk-free rate remains close to zero for most of the sample period and exhibits only limited variation, while the market return shows substantially higher volatility.

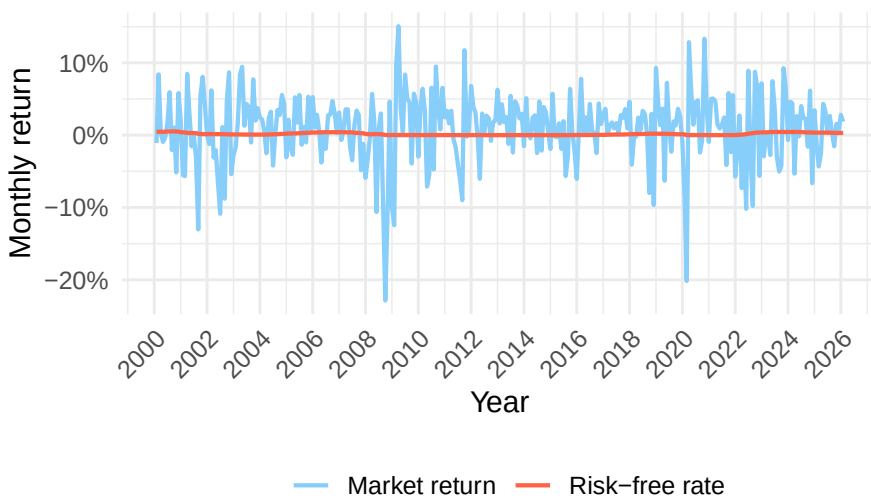

Figure 1: Time Series of Market Return and Risk-Free Rate

Table 1 reports summary statistics of monthly stock returns. The mean return is close to zero, while the dispersion is substantial, as indicated by the standard deviation.

Table 2: Summary statistics of estimated CAPM beta coefficients

Mean	Variance	Minimum	Maximum	Significant (5%) [%]	Significant (1%) [%]
1.02	0.168	0.179	2.76	100	98.915

1.3 Estimate of betta coefficients of the characteristic curves

For each stock, the coefficients of the characteristic curve are estimated using the following model:

$$r_j - r_f = \alpha_j + \beta_j(r_m - r_f) + \varepsilon_j \tag{3}$$

The CAPM beta coefficients are estimated using time-series regressions of excess stock returns on excess market returns, revealing substantial variation in systematic risk across assets.

As shown in Table 2, the average β is approximately 1.02, indicating that, on average, stocks exhibit similar sensitivity to market movements as the market portfolio. However, individual β values vary considerably, ranging from 0.179 to 2.76. Some stocks therefore display low systematic risk ($\beta < 1$), while others exhibit high sensitivity ($\beta > 1$).

As reported in Table 2, all estimated β coefficients are statistically significant at the 5% level, and 98.915% remain significant at the 1% level, providing strong evidence that market returns explain individual stock returns.

The Security Characteristic Lines of selected stocks are shown in Figure 2.

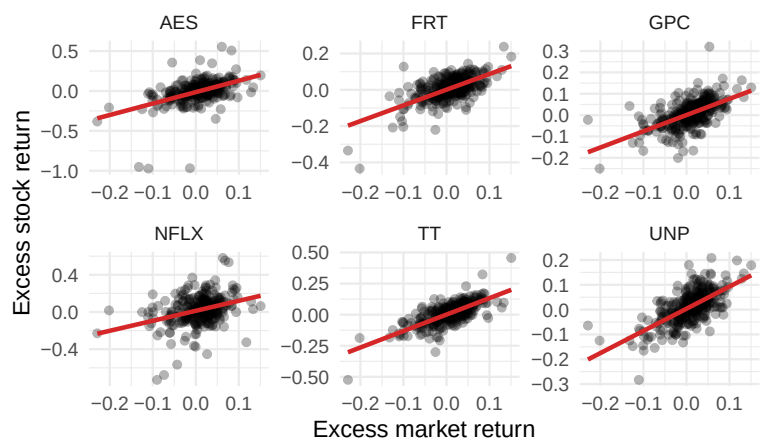

Figure 2: Security Characteristic Lines of selected stocks

1.4 Testing hypotheses on the intercept of the CAPM model

For each stock, we test the hypothesis that the intercept (alpha) in the CAPM model is equal to zero:

$$H_0 : \alpha_j = 0 \quad \text{vs.} \quad H_1 : \alpha_j \neq 0 \tag{4}$$

The tests are conducted at the 5% significance level using t-tests based on the estimated regression coefficients.

As shown in Table 3, the null hypothesis is rejected for only a small fraction of stocks (8.46%), while it is not rejected for the majority (91.54%). This suggests that, for most assets, the CAPM intercept is not statistically different from zero. This means that, for most assets, there is no evidence that they generate returns beside what would be expected after accounting for market risk.

Table 3: CAPM intercept significance (5% level)

Significant on 5%	count	percentage
FALSE	422	91.54
TRUE	39	8.46

Table 4: Security Market Line regression results

Term	Estimate	Std. Error	t-statistic	p-value
γ_0	0.0100	0.001	9.5225	0.0000
γ_1	-0.0023	0.001	-2.3973	0.0276

2 Testing CAPM Validity

2.1 Portfolio Formation and Beta Aggregation

Stocks are sorted by their estimated β coefficients and assigned to 20 portfolios of roughly equal size. Each portfolio is equally weighted, so the portfolio beta is the arithmetic mean of individual betas. Portfolio excess returns are computed as the time-series average of equally weighted stock excess returns over the full sample period.

Figure 3 reveals a pattern inconsistent with CAPM - mean excess returns do not increase systematically with beta. Instead, the relationship appears hump-shaped rather than linear, suggesting a possible quadratic structure, which is tested formally in Section 2.3.

2.2 Security Market Line

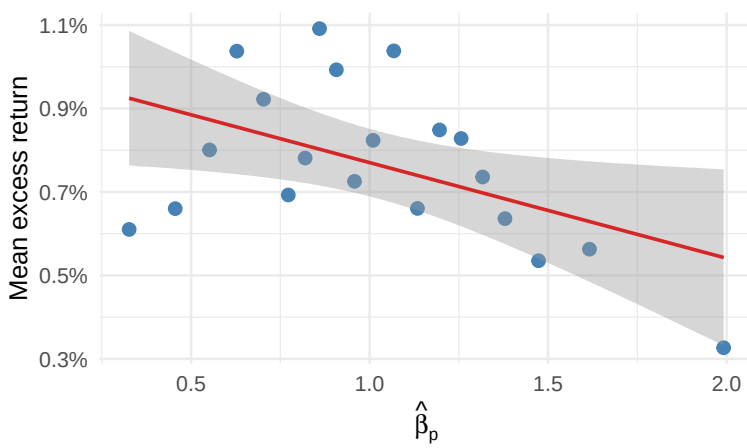

Figure 3: Security Market Line estimated from portfolio data

The Security Market Line (SML) is estimated by regressing mean portfolio excess returns on portfolio betas:

$$r_{p_i} - r_f = \gamma_0 + \gamma_1 \beta_{p_i} + \varepsilon_{p_i} \tag{5}$$

The estimated $\hat{\gamma}_0 = 0.01$ (statistically significant at 5%) and $\hat{\gamma}_1 =$ -0.0023 (statistically significant at 5%). The observed mean market risk premium over the sample period is 0.0078.

Table 5: Extended CAPM specification (Fama-MacBeth/Gibbons)

Term	Estimate	Std. Error	t-statistic	p-value
δ_1	0.0048	0.0019	2.5950	0.0195
δ_2	0.0091	0.0033	2.7538	0.0141
δ_3	-0.0048	0.0016	-3.0088	0.0083
δ_4	-0.0967	0.2307	-0.4194	0.6805

CAPM imposes two testable restrictions on the SML. First, γ_0 should be zero, since if portfolio excess returns are fully explained by systematic risk, no intercept should remain. Second, γ_1 should equal the observed market risk premium $\bar{r}_m - r_f$.

The intercept $\hat{\gamma}_0$ is statistically significant at the 5% level, which rejects the first CAPM restriction — some component of portfolio returns is not explained by market risk alone.

To test the second restriction, we assess whether $\gamma_1 = \bar{r}_m - r_f$ using a t-test:

$$t = \frac{\hat{\gamma}_1 - (\bar{r}_m - r_f)}{se(\hat{\gamma}_1)} \tag{6}$$

The test yields t = -10.545 (p-value < 0.0005), so the hypothesis $\gamma_1 = \bar{r}_m - r_f$ is rejected at the 5% level. The estimated security market line exhibits a negative slope, indicating that higher-beta portfolios do not earn higher average returns in this sample. This contradicts the standard CAPM prediction and suggests that beta alone does not adequately explain the cross-sectional variation in returns

2.3 Extended CAPM Specification

Following Fama and MacBeth (1973) and Gibbons (1982), we test whether the linear CAPM specification is adequate by adding a squared beta term and a measure of idiosyncratic risk:

$$\bar{r}_p - r_f = \delta_1 + \delta_2 \hat{\beta}_p + \delta_3 \hat{\beta}_p^2 + \delta_4 \hat{\sigma}^2_{\varepsilon_p} + \varepsilon_p \tag{7}$$

If CAPM holds, δ_3 and δ_4 should both be zero: $\delta_3 = 0$ tests for linearity, $\delta_4 = 0$ tests whether idiosyncratic risk is priced.

Table 5 reports the estimates. At the 5% significance level, δ_2 is positive and statistically significant (p-value = 0.0141), indicating a positive relationship between beta and returns. The coefficient δ_3 is also statistically significant (p-value = 0.0083), suggesting a non-linear relationship and rejecting the linearity assumption of CAPM. In contrast, δ_4 is not statistically significant (p-value = 0.6805), indicating that idiosyncratic risk is not priced. The intercept δ_1 is statistically significant (p-value = 0.0195), implying that average returns are not fully explained by beta.

Overall, the results provide only partial support for CAPM. The pricing of idiosyncratic risk is not supported, but the linearity assumption is violated and the intercept is significantly different from zero

3 Structural Breaks and Asymmetry in CAPM

3.1 Chow Tests on Individual Characteristic Curves

The CAPM characteristic curve is reformulated to allow for a change in β over the last 48 months of the sample:

$$r_j - r_f = \alpha_j + \beta_j (r_m - r_f) + \delta_j \cdot D_t \cdot (r_m - r_f) + \varepsilon_j \tag{8}$$

Table 6: Proportion of stocks with a significant structural break (%)

Chow break — sig. at 5% [%]	Chow break — sig. at 1% [%]	Chow forecast — sig. at 5% [%]	Chow forecast — sig. at 1% [%]
13.88	4.56	14.1	9.11

Table 7: SML estimates by subperiod

Period	Term	Estimate	Std. Error	t-stat	p-value
early	γ_0	0.0115	0.0010	12.0833	0.0000
early	γ_1	-0.0031	0.0008	-3.6171	0.0020
recent	γ_0	0.0029	0.0015	1.9681	0.0647
recent	γ_1	0.0001	0.0014	0.0411	0.9677

where $D_t = 1$ if t falls in the last 48 months, and zero otherwise. Two variants of the Chow test are applied to each stock:

- **Chow structural break test**: tests whether all coefficients are identical across the two subperiods, using the full-sample residual sum of squares and the sum from two separate regressions.
- **Chow forecast (predictive) test**: tests whether the model estimated on the early period predicts returns in the recent period.

At the 5% significance level, the Chow structural break test detects instability in about 13.9% of stocks, while at the 1% level this share falls to 4.6%. The Chow forecast test yields rejection rates of 14.1% at the 5% level and 9.1% at the 1% level.

Overall, most stocks do not show evidence of instability, but a non-negligible share does. Rejection rates are slightly higher in the forecast test, suggesting that out-of-sample fit is weaker than what parameter stability alone would indicate.

3.2 Security Market Line by Subperiod

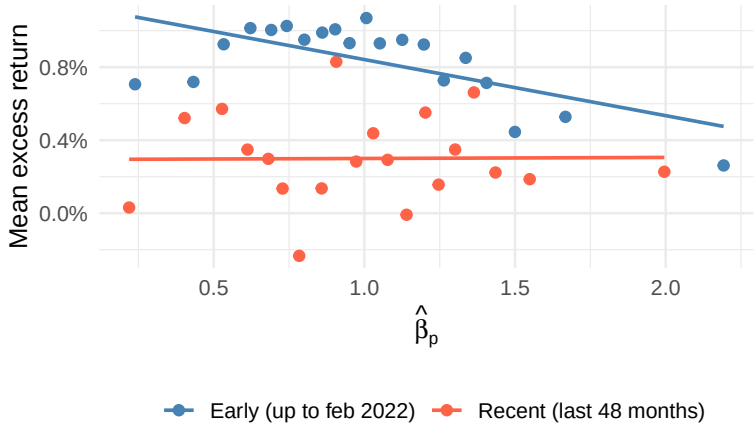

Figure 4: Security Market Line: early vs. recent subperiod

Table 7 and Figure 4 show the SML estimated separately for the early and recent subperiods.

At the 5% significance level, in the early period the intercept (γ_0) is positive and statistically significant (p-value = 0.0000), while the slope (γ_1) is negative and statistically significant (p-value = 0.0020). This implies a downward-sloping SML and a negative relationship between beta and returns, which contradicts the CAPM prediction of a positive risk–return trade-off.

Table 8: Asymmetric beta estimates: summary

Mean beta+	Mean beta-	Mean diff	Sig. diff at 5% [%]
1.018	1.021	-0.003	17.57

Table 9: Test of $\gamma_1 = \bar{r}_m - r_f$ by market state

Market state	$\hat{\gamma}_1$	$\bar{r}_m - r_f$	t-stat	p-value
Down	-0.0396	-0.0391	-0.3336	0.7425
Up	0.0226	0.0362	-15.6566	0.0000

In the recent period, the intercept is not statistically significant at the 5% level (p-value = 0.0647), and the slope is also not significant (p-value = 0.9677), indicating no meaningful relationship between beta and returns.

Overall, the estimated SMLs do not support the CAPM in either subperiod.

3.3 Asymmetric Beta

The characteristic curve is reformulated to allow β to differ depending on whether the market return is positive or negative:

$$r_j - r_f = \alpha_j + \beta_j^+ (r_m - r_f)^+ + \beta_j^- (r_m - r_f)^- + \varepsilon_j \tag{9}$$

where $(r_m - r_f)^+ = \max(r_m - r_f, 0)$ and $(r_m - r_f)^- = \min(r_m - r_f, 0)$.

For each stock, we test $H_0 : \beta^+ = \beta^-$ against $H_1 : \beta^+ \neq \beta^-$ at the 5% significance level using a t-test based on the asymptotic standard error of the difference.

Table 8 reports the results of the asymmetric beta estimates. The mean estimates of β^+ and β^- are nearly identical (1.018 and 1.021), resulting in an average difference close to zero (-0.003). This indicates no meaningful asymmetry in beta at the aggregate level.

At the individual stock level, 17.57% of stocks show a statistically significant difference between β^+ and β^- at the 5% level. This suggests that asymmetry may be present for some stocks, although it is not consistently observed across the sample. [1]

Overall, the evidence for asymmetry is mixed: while some individual stocks show significant differences, the aggregate results do not provide strong or consistent evidence of systematic asymmetry in beta.

3.4 Security Market Line in Up and Down Markets

Figure 5 and Table 10 show the SML estimates across market regimes, while Table 9 reports the test of $\gamma_1 = \bar{r}_m - r_f$.

In down markets, the slope (γ_1) is negative and highly significant, indicating that higher-beta portfolios experience larger losses. The intercept (γ_0) is not statistically different from zero. The equality $\gamma_1 = \bar{r}_m - r_f$ is not rejected (p = 0.743), so the slope is consistent with the observed negative market premium.

[1] We also considered an alternative specification of the characteristic curve without the intercept. In this case, 8.5% of stocks exhibit a statistically significant difference between β^+ and β^- at the 5% significance level. The mean difference between β^+ and β^- was 0.001, which is close to zero.

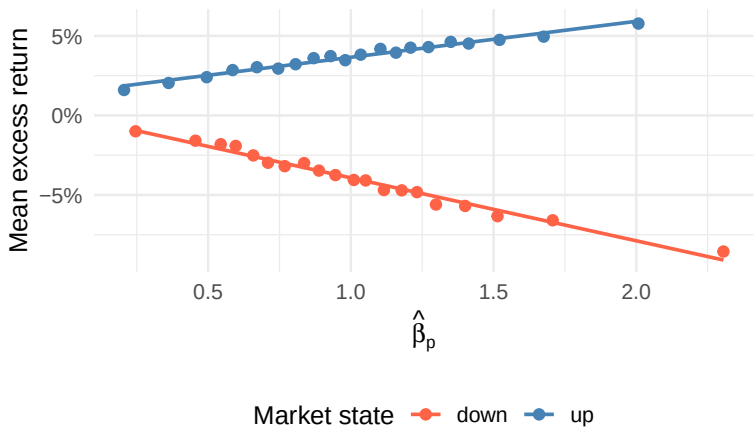

Figure 5: Security Market Line: up vs. down market months

Table 10: SML estimates: up vs. down market months

Market state	Term	Estimate	Std. Error	t-stat	p-value
down	γ_0	0.0003	0.0014	0.2026	0.8418
down	γ_1	-0.0396	0.0013	-30.7565	0.0000
up	γ_0	0.0139	0.0010	14.4298	0.0000
up	γ_1	0.0226	0.0009	25.9906	0.0000

In up markets, the slope is positive and significant but smaller than the market premium, and the equality $\gamma_1 = \bar{r}_m - r_f$ is strongly rejected ($p < 0.001$). The intercept is positive and significant, indicating that returns are not fully explained by beta.

In down markets, the CAPM restrictions are not rejected, as the slope is consistent with the market risk premium and the intercept is not statistically different from zero. In contrast, in up markets, the intercept is significantly different from zero and the slope is statistically not equal to the market risk premium, indicating a violation of the CAPM.